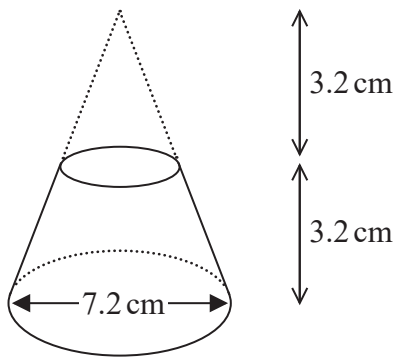


20 Here is a frustum of a cone.

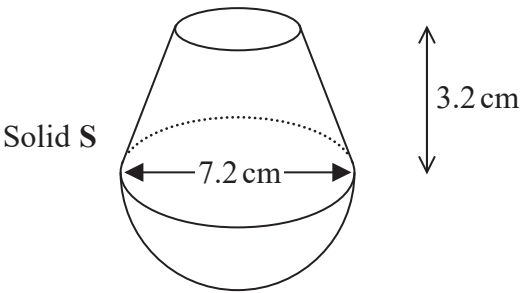


Volume of sphere =  $\frac{4}{3}\pi r^3$

Volume of cone =  $\frac{1}{3}\pi r^2 h$

The diagram shows that the frustum is made by removing a cone with height 3.2 cm from a solid cone with height 6.4 cm and base diameter 7.2 cm.

The frustum is joined to a solid hemisphere of diameter 7.2 cm to form the solid S shown below.



The density of the frustum is 2.4 g/cm<sup>3</sup>  
The density of the hemisphere is 4.8 g/cm<sup>3</sup>

Calculate the average density of solid S.

.....g/cm<sup>3</sup>